Further Strategies — problem 14.53

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Let $A = (a_1, a_2, \ldots, a_m)$ with $m = 2^n$ and $a_i \in \{-1, +1\}$. Consider the transformation $T(A) = (a_1a_2, a_2a_3, \ldots a_ma_1)$. Prove that, by repeated application of this transformation, you will reach the m-tuple $(1, 1, \ldots, 1)$.